DAY - 3

ARRAY 1D & 2D

Imagine you are storing marks of 100 students

Bad way

int m1, m2, m3, m4 - - -

Smart way

Put all related data in one container.

That container is an array.

An array is like:

* A row of numbered lockers
* Each locker stores one value
* you access lockers using index numbers

1. What is an Array?

An array stores multiple values of same type in continuous memory.

```
int[] marks = new int[5];
```

means

* int[] → array of integers
* marks → array name
* 5 → Number of lockers

index: starts from 0 - 4

index always starts from 0

This is not random — it's how memory works internally

WHY index starts from 0

memory is like:

Base Address + off set

index 0 = base address

index 1 = base + 1 block

* Starting from 0 makes calculations faster and simpler for CPU.

Creating and using Array

Ex: Store and print marks

```
int [] marks = {85, 90, 78, 92, 88};

for (int i=0; i< marks.length; i++) {
    System.out.println(marks[i]);
}
```

what happens?

marks.length → total lockers

i → current locker number

mark[i] → Value inside locker

Loop + Array = Best Friends

DSA Logic

Human thinking

compare everything and remember the biggest.

code:

```
int[] arr = {10, 25, 5, 40, 30};
int max = arr[0];

for (int i = 1; i<arr.length; i++) {
    if (arr[i] > max) {
        max = arr[i];
    }
}

System.out.println("max:" + max);
```

How it works

* Assume first is biggest
* update when you find better
* This pattern repeats everywhere in DAS

2D Array

Mental model:

Excel sheet = rows + columns.

```
int[][] matrix = {
    {1, 2, 3},
    {4, 5, 6}
};
```

Access

matrix [row][column]

Ex:

```
int [][] mat = {
    {1, 2},
    {3, 4},
};
for (int i = 0; i < mat.length ; i++) {
    for (int j = 0; j < mat[i].length; j++) {
        System.out.println (mat[i][j] + " ");
    }
    System.out.println ();
}
```

Key points

* Array stores same-type data
* index starts from 0
* length gives size
* Loop used to travers
* 2D array = table structure.